

CCST9012 Tutorial Week 5

3 Artificial Constructions

1 THREE ARTIFICIAL CONSTRUCTIONS

In the following exercise, we will briefly explore the function of the three artificial constructions used by Ptolemy in his models of planetary motions.

1.1 EPICYCLE

There are two important parameters here:

1. Epicycle Period: The time it takes the Planet to complete a full rotation around the epicycle divided by the time it takes the epicycle to complete a full rotation around the deferent.
2. Epicycle Size: The radius of the epicycle divided by the radius of the deferent.

The mathematics to prove this is a bit complicated. Demonstrate that the epicycle will create a retrograde motion whenever the Epicycle Period is less than the Epicycle Radius.

[**Bonus:** The planet has two components of speed, one from the motion of the epicycle on the deferent and one from the motion of the epicycle itself. Let v_d be the speed of the epicycle on the deferent. Let r_d be the radius of the deferent and T_d be its period. The epicycle travels at a speed given by

$$v_d = \frac{2 \pi r_d}{T_d},$$

completing one circumference of the deferent every period.

Similarly, we can let v_e be the speed of the planet on the epicycle. It will be given by

$$v_e = \frac{2 \pi r_e}{T_e},$$

where r_e and T_e are the epicycles radius and period respectively.

For retrograde motion to occur, the epicycle speed must be greater than the deferent speed, $v_e > v_d$ or

$$\frac{2 \pi r_e}{T_e} > \frac{2 \pi r_d}{T_d}.$$

This can be rearranged to show that for retrograde motion to occur,

$$\frac{r_e}{r_d} > \frac{T_e}{T_d}.$$

We characterize the motion of a planet by its tropical period and its synodic period.

- The tropical period is the average time it takes for the planet to complete a full cycle through the ecliptic. This is the same as the time it takes the epicycle to complete a full rotation around the deferent.
- The synodic period is the time between retrograde motions. This is related to, but not equivalent to the epicycle period.

Set the epicycle period to 0.6 (and the epicycle size to any value larger than 0.6). How often do retrograde motions occur.

In this case, we see that a retrograde motion occurs every 1.5 tropical periods.

Answer the same for the epicycle period=0.5.

In this case, we see that a retrograde motion occurs exactly once every tropical period.

[**Bonus:** Mathematically inclined students looking for exercise may show that the synodic period is given by,

$$T_s = \frac{T_e T_d}{T_d - T_e}.$$

Since the synodic and deferent periods can be observed, the period of the epicycle's motion can be inferred through observations.]

1.2 ECCENTRIC AND EQUANT

The purpose of the Eccentric and Equant constructions, is to account for variation in the apparent speed of planets. Their effect on the apparent motion is qualitatively the same, but in detail they are slightly different.

Placing the Earth far from the Center of the Deferent, 'O' will exaggerate the effect of the Eccentric construction. For the Equant, placing the Center of Motion, 'CoM', far from the earth will exaggerate its effect.

For each of these constructions, adjust them such that the motion along the ecliptic, as viewed from Earth will be fastest near the Summer Solstice. Repeat this for two other constellations.

For both constructions, place 'O' on the opposite side of the Earth from the constellation where you want the planet to move the fastest.

True or False: In a model using only the eccentric, the planet moves along the deferent at constant speed. In a model using only the equant, the planet moves along the deferent at non-uniform speed.

This is true. In the first case, the object moves at a constant speed along but because its distance from earth changes, its apparent speed is faster when its closer to earth and slower when its further away. In the second case, the apparent speed as seen from earth is the same as the speed of along the deferent because in this case, the distance between the planet and the Earth is always the same. The planet speeds up when its further from the Center of Motion and slows down when its closer.

Place 'O' in the Eccentric Construction in the same position as 'CoM' in the Equant Construction. Make sure both have the same 'Tropical Period' value. Compare the number of days in each season to see that the effects of these two constructions are not identical.

To recreate these numbers, keep the Tropical Period at 500. Place 'O' directly between Earth and the Autumnal Equinox, two units away on the grid. Start each with angle on the far left of the slider so that the planet is seen at the Vernal Equinox from Earth. This corresponds to tropical cycle time =0

	VE	SS	AE	WS
Eccentric	0	51.67	250	448.33
Equant	0	71.67	250	428.33